Katherine Lee

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EDUCATION

Ph.D. in Astronomy, Uppsala University, expected Apr 2030
Master in Astronomy, University of Oslo, June 2023
Bachelor of Science in Physics, cum laude, Montana State University, May 2020

PUBLICATIONS

V. Ksoll **et al.** A method to derive self-consistent NLTE astrophysical parameters for four million high-resolution 4MOST stellar spectra in half a day with invertible neural networks. A&A, Apr 2026. https://doi.org/10.1051/0004-6361/202558595

H. Rauer **et al.** The PLATO Mission. Experimental Astronomy, June 2025. https://doi.org/10.1007/s10686-025-09985-9

D. J. Watts, U. Fuskeland, **et al.** Cosmoglobe DR1 results. III. First full-sky model of polarized synchrotron emission from all WMAP and Planck LFI data. A&A, June 2024. https://doi.org/10.1051/0004-6361/202348330

J. R. Eskilt, D. J. Watts, **et al.** Cosmoglobe DR1 results. II. Constraints on isotropic cosmic birefringence from reprocessed WMAP and Planck LFI data. A&A, Nov 2023. https://doi.org/10.1051/0004-6361/202346829

D. J. Watts **et al.** Cosmoglobe DR1 results. I. Improved Wilkinson Microwave Anisotropy Probe maps through Bayesian end-to-end analysis. A&A, Nov 2023. https://doi.org/10.1051/0004-6361/202346414

J. R. Eskilt, **K. Lee**, D. J. Watts, et al. Cosmoglobe: Towards end-to-end CMB cosmological parameter estimation without likelihood approximations. A&A, Oct 2023. https://doi.org/10.1051/0004-6361/202347358

RESEARCH

Ph.D., Uppsala University — Jan 2026 - present

Investigating the nature of Population III stars by chemically characterising ultra metal-poor stars in the Milky Way, with a focus on measuring the abundances of oxygen and carbon in order to constrain the primordial initial mass function. Supervised by Thomas Nordlander.

SAPP head developer, Max Planck Institute for Astronomy — Oct 2023 - Sep 2025

Head developer for Stellar Abundances and atmospheric Parameters Pipeline (SAPP, Gent et al. 2022; Lee et al., in prep), which performs a Bayesian analysis of stellar spectroscopic, photometric, interferometric, and asteroseismic data to estimate parameters including effective temperature, surface gravity, metallicity, and chemical abundances. Developed SAPP as part of the PLATO and 4MOST pipelines, maintaining and upgrading three separate versions in total. Advised two bachelor's students as they used SAPP to complete their theses. Used SAPP to analyse 4MOST target stars for galactic archeology; a paper about this work is in preparation. Supervised by Maria Bergemann.

Master thesis, University of Oslo — June 2022 - June 2023

Worked on cosmological parameter estimation using Commander, which processes CMB data using Bayesian Gibbs sampling to produce a joint end-to-end analysis from raw data to cosmological parameters. Integrated ΛCDM parameter sampling into Commander, and prepared a reanalysis of COBE-FIRAS data to produce better estimates of CMB spectral distortions. Supervised by Duncan Watts, Hans Kristian Eriksen, Ingunn Wehus, and Johannes Røsok Eskilt.

Exoplanet catalog generation, NASA GSFC — Jan - Apr 2021

Created a catalog of potential exoplanet direct imaging targets, meant primarily for use with JWST. Retrieved stellar data from a wide variety of databases, characterised each entry using astrometric, photometric, and asteroseismic properties, and designed an algorithm to rank each star according to its usefulness as a target. Supervised by Eliad Peretz and John Mather.

Sodium lidar experiment design, NASA GSFC — June - Aug 2019

Produced a proof of concept for a proposed balloon-borne sodium lidar. Designed an optical system using modelling software packages Zemax and TracePro to maximise the mapping potential of the instrument, and built and tested this system in the laboratory. Presented a poster at the APS April Meeting in 2020. Supervised by Michael Krainak.

Image correction, National Solar Observatory — May - Aug 2018

Wrote a program to correct the effects of atmospheric interference on images taken with the Daniel K. Inouye Solar Telescope. The program was adapted to Python based on existing routines in IDL, and corrected for atmospheric dispersion and turbulence. Supervised by Kevin Reardon.

Eclipse Ballooning Project, Montana Space Grant — 2015 - 2017

As part of the BOREALIS high-altitude ballooning team, helped lead a nationwide outreach effort in which over 50 teams live-streamed the August 2017 solar eclipse from high-altitude balloons. Flew a Raspberry Pi camera to image the eclipse from Earth's upper atmosphere. Supervised by Angela Des Jardins and Berk Knighton.

CME and solar flare analysis, Montana State University — June 2014 - Dec 2016

Created a catalogue of high-energy (M- and X-class) solar flares from the RHESSI and GOES datasets. Tracked notable coronal mass ejections in IDL using images from STEREO and SDO in order to establish their trajectories over time and their associations with corresponding flares. Supervised by Angela Des Jardins.

OTHER EXPERIENCE

Astrobites author — Jan 2023 - present

I write summaries of recent papers in astronomy geared towards an undergraduate or non-specialist audience as part of the Astrobites collaboration. While I reached the end of my term as an active author in Feb 2025, I'm still active on the climate change, diversity, editorial, and recruitment committees. My articles can be found here.

Skype a Scientist — 2022 - present

I volunteer for Skype a Scientist, where teachers can sign up for a virtual visit from a scientist in a particular field of interest. I've given talks and answered questions for students from US third to eleventh grades on everything from lasers to exoplanets to general relativity.

Student government, University of Oslo — Aug 2022 - June 2023

Represented the astronomy master' students in the department of mathematics and natural sciences at UiO. Worked to improve the study environment as part of astronomy department student government.

PROGRAMMING LANGUAGES

- Python
- Fortran
- MATLAB
- C++
- ADQL
- IDL
- Unix/Linux
- R
- HTML

LANGUAGES

English — Native
Norwegian — C1
Spanish — B2 reading, B1 speaking/writing
German — B1